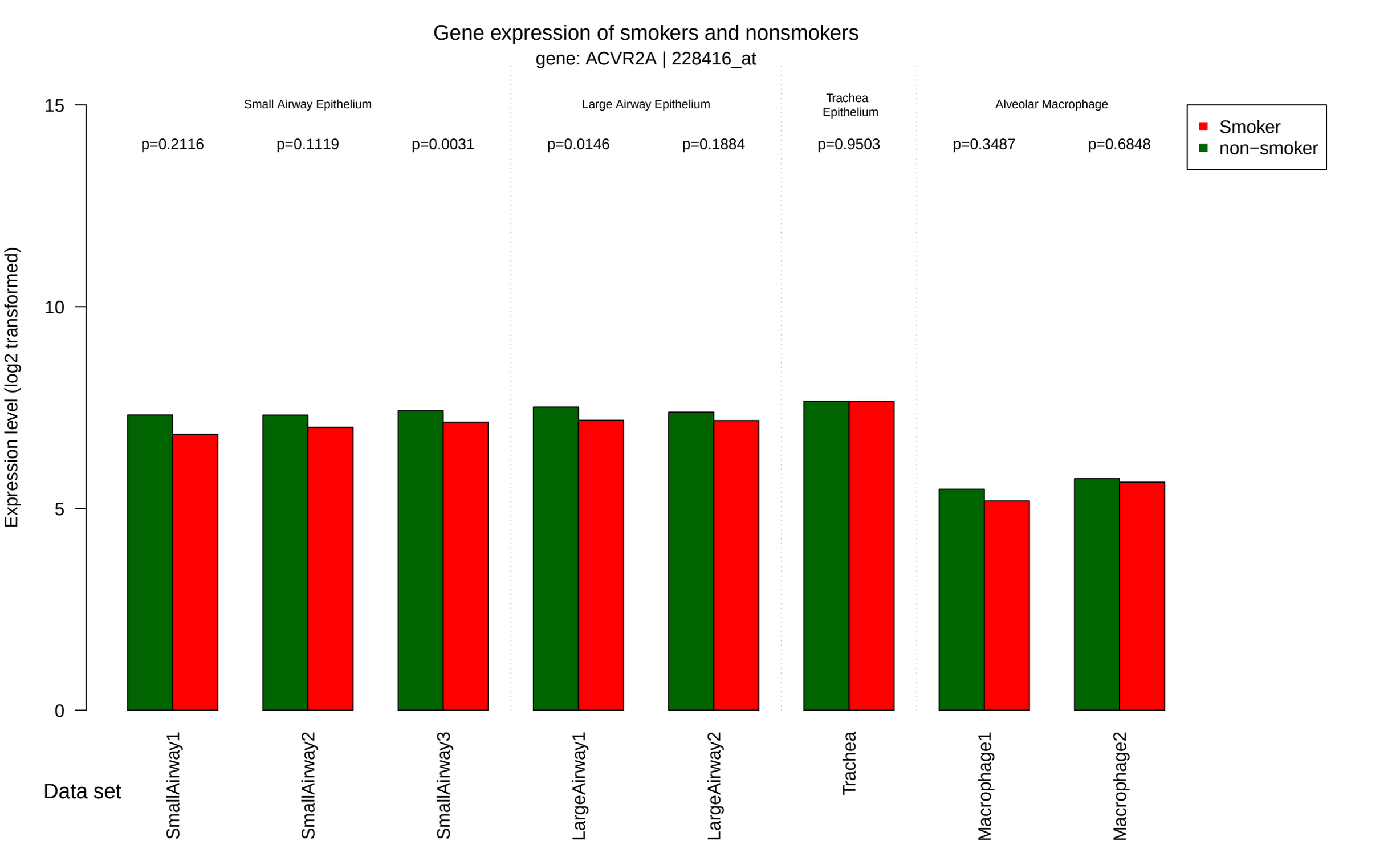
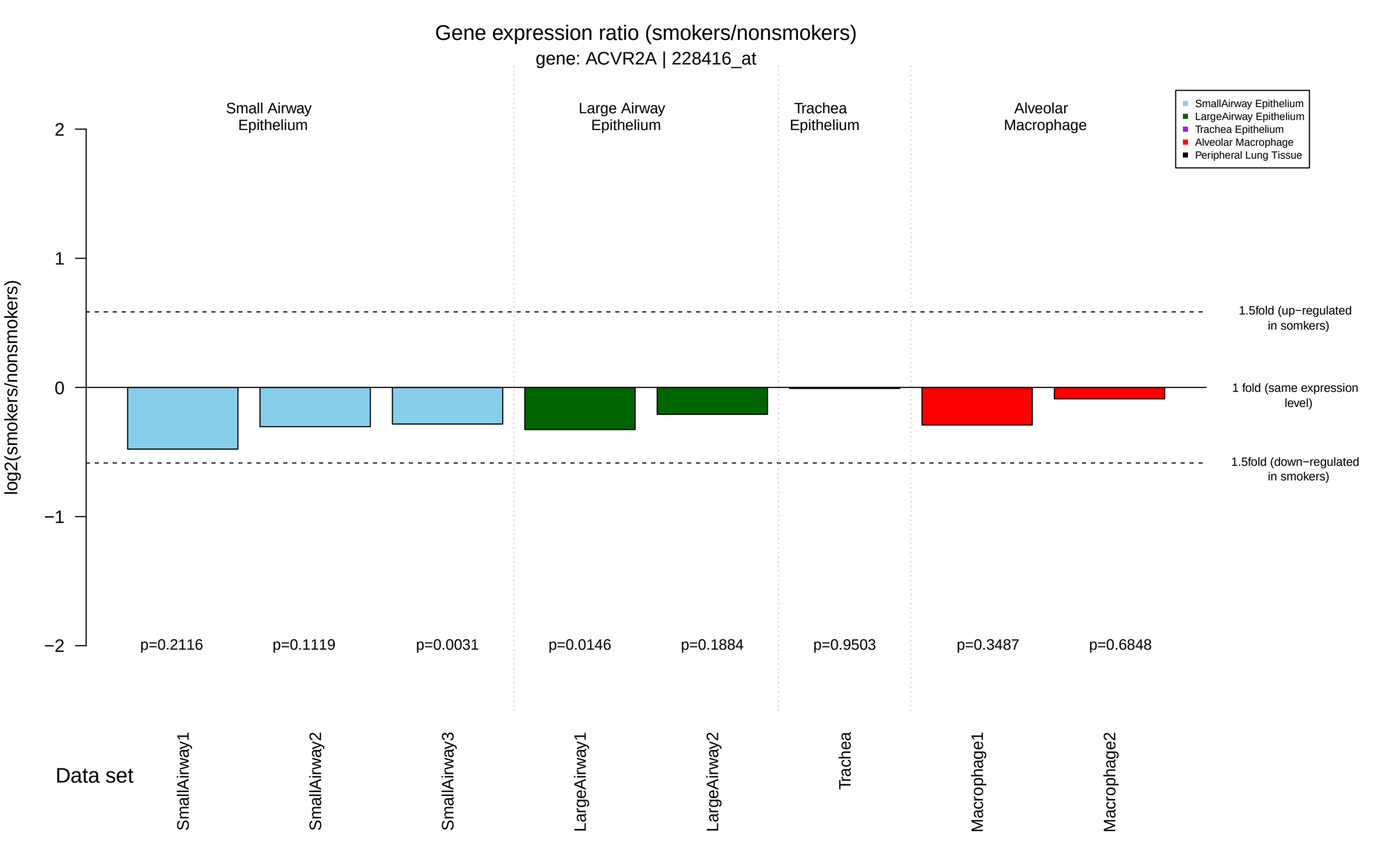
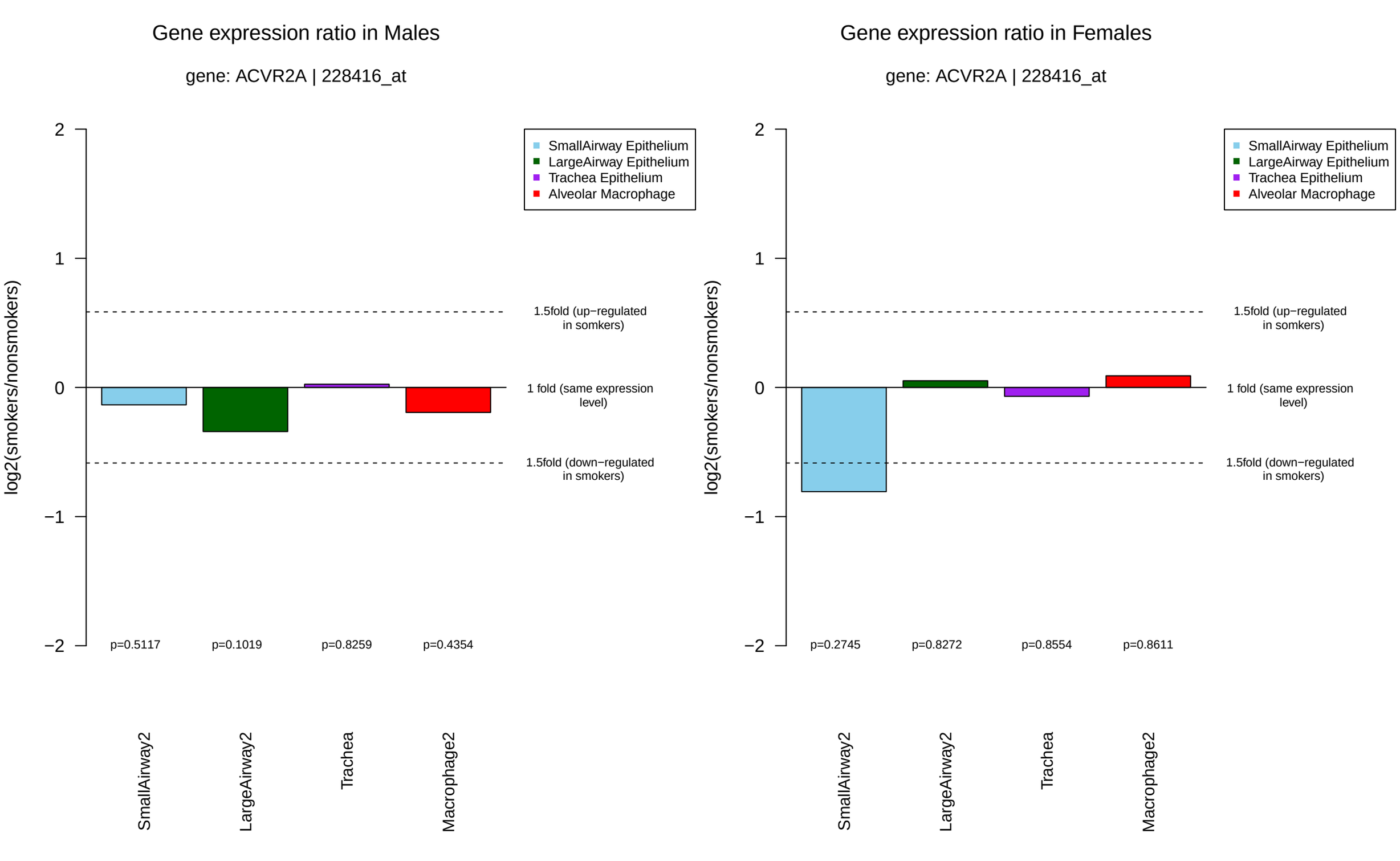
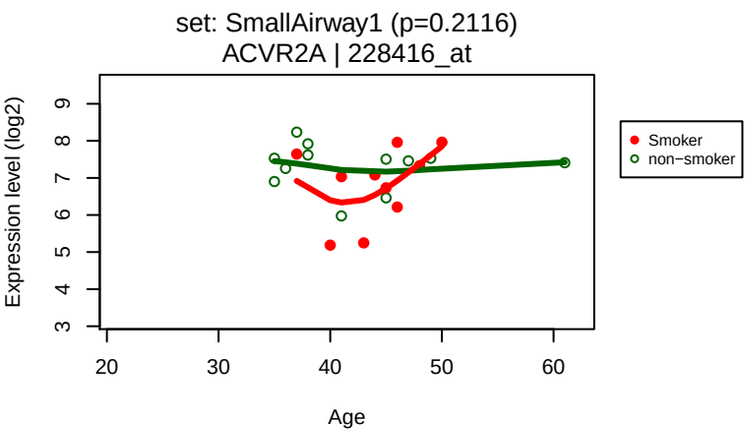
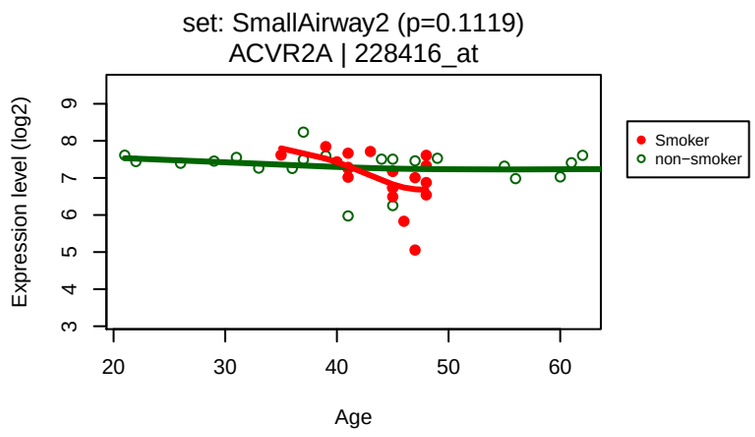
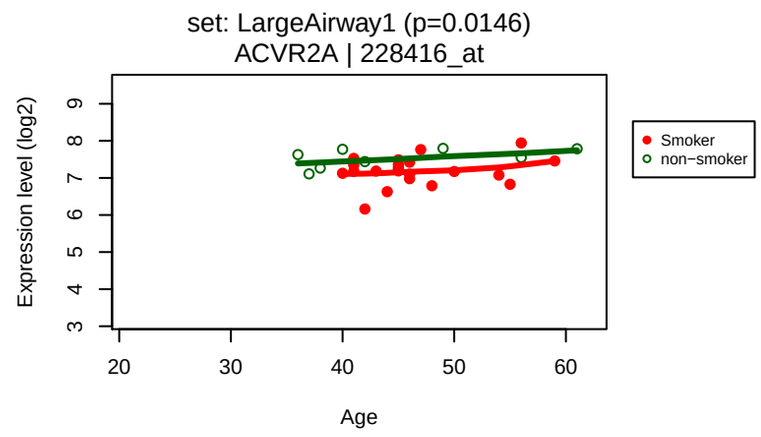

Gene expression of smokers and nonsmokers
gene: ACVR2A | 228416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2116
p=0.1119
p=0.0031
p=0.0146
p=0.1884
p=0.9503
p=0.3487
p=0.6848
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

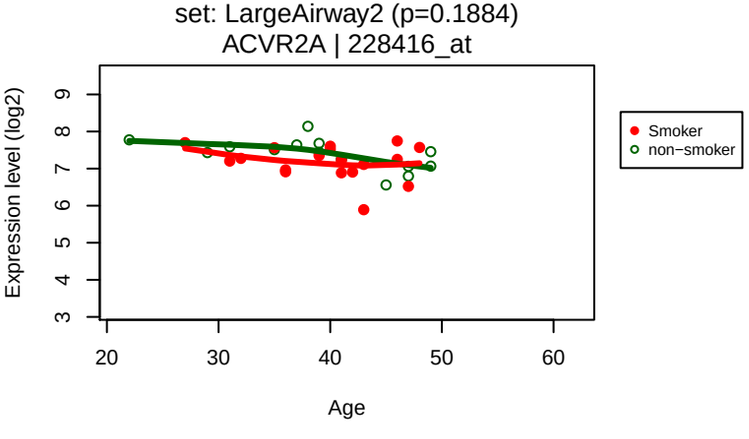

Gene expression ratio (smokers/nonsmokers)
gene: ACVR2A | 228416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2116
p=0.1119
p=0.0031
p=0.0146
p=0.1884
p=0.9503
p=0.3487
p=0.6848
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

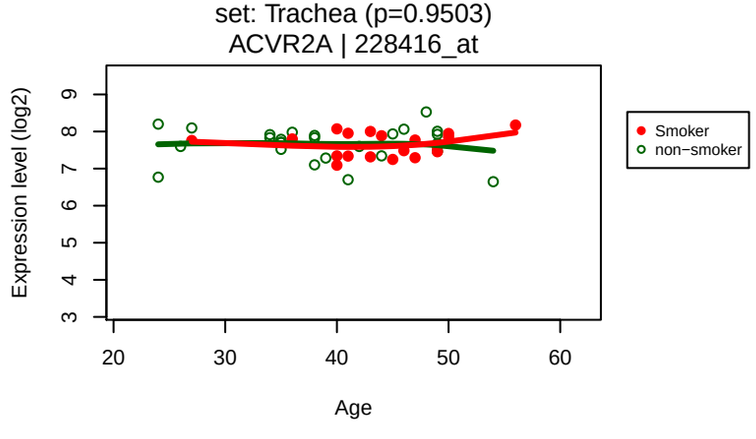

Gene expression ratio in Males
gene: ACVR2A | 228416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5117
p=0.1019
p=0.8259
p=0.4354
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACVR2A | 228416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2745
p=0.8272
p=0.8554
p=0.8611
SmallAirway2
LargeAirway2
Trachea
Macrophage2

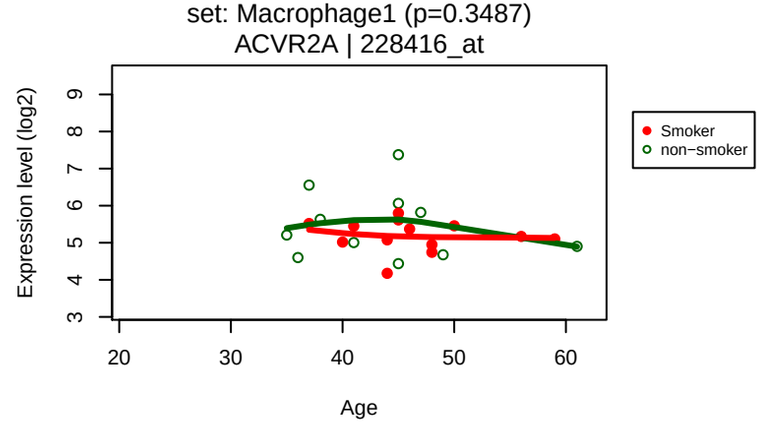

set: SmallAirway1 (p=0.2116)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker

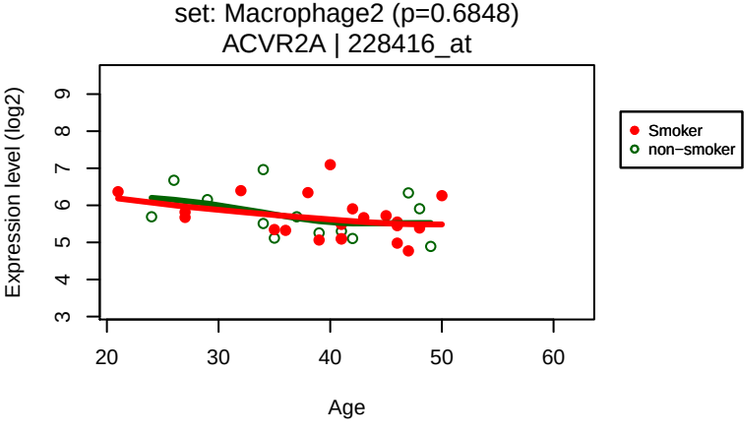

set: SmallAirway2 (p=0.1119)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker

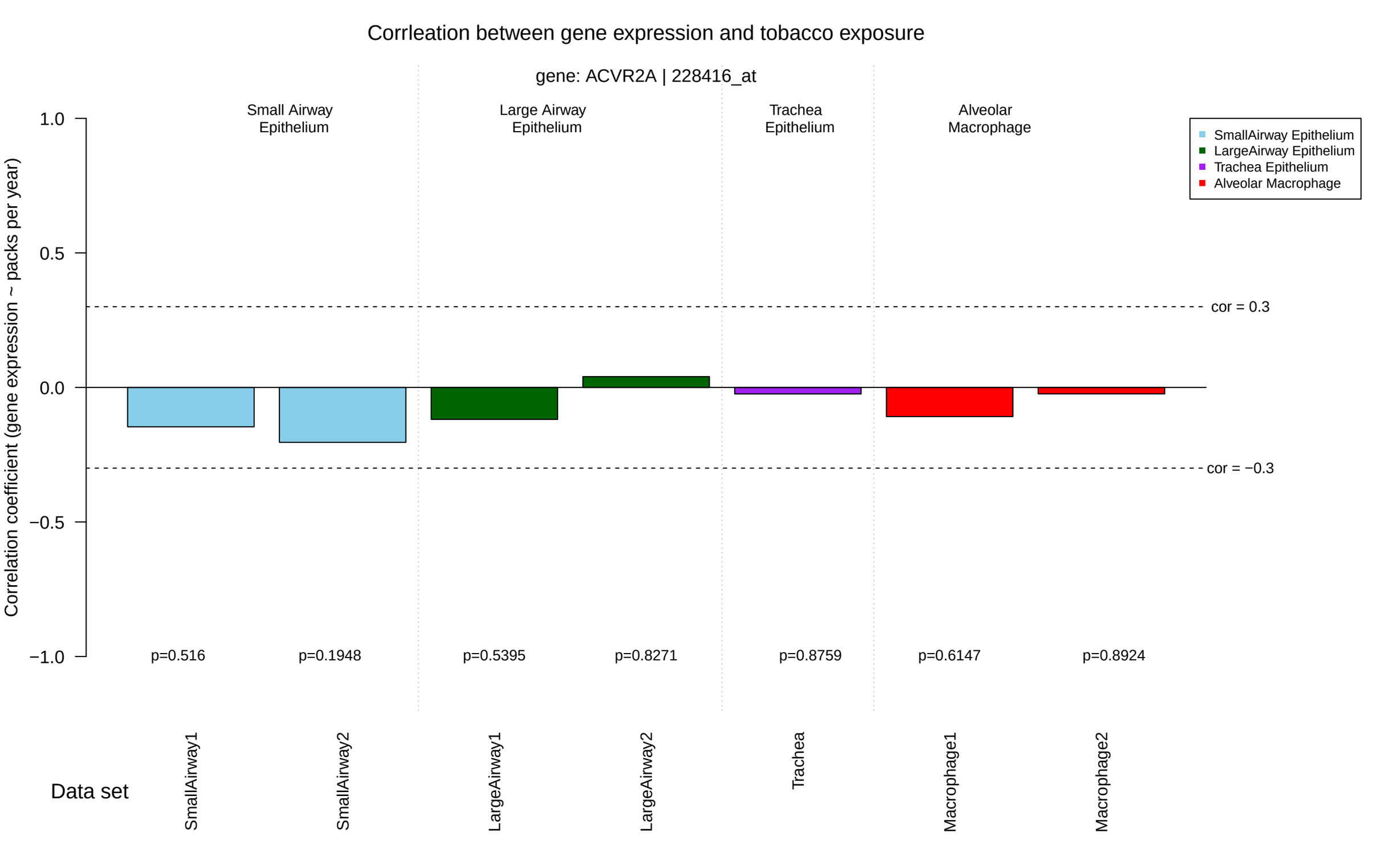

set: LargeAirway1 (p=0.0146)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1884)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9503)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3487)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6848)
ACVR2A | 228416_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACVR2A | 228416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.516
p=0.1948
p=0.5395
p=0.8271
p=0.8759
p=0.6147
p=0.8924
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2